

MIRA — Mandala-Inspired Memory for AI Retrieval

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Reproducibility statement

Every quantitative claim in this paper is generated from committed run artifacts by `paper/export_results.py --real` ; no table is hand-edited. All experiments run on a single consumer laptop (RTX 3050 4GB, 16 GB RAM) with zero cloud calls. Artifacts and builders:

- `data_bench/bench_real_results.json`
- `data_bench/bench_real_answers.json`
- `data_bench/ablation_real_results.json`
- `data_bench/scale_sweep_results.json`
- `data_bench/hotpotqa_bench_real_results.json`
- `data_indic/indicqa_results.json`
- `data_indic_ml/indicqa_ml_results.json`
- `experiments/neural_validation.json`

Abstract

Retrieval-augmented generation (RAG) systems predominantly organize memory as a flat vector collection, optionally augmented by entity graphs or hierarchy. We investigate an underexplored structural alternative: a **radial memory topology** in which memories are organized in concentric rings around a core concept, partitioned into semantic sectors, and positioned by a formally defined radial distance combining semantic, hierarchical, graph, and temporal signals. We implement MIRA, a local-first research platform in which the radial mechanism is one switchable component among nine, each independently ablatable, evaluated against naive vector RAG, hybrid vector+graph RAG, and hierarchical retrieval under a declared, shared protocol when the answer-stage experiment is run; the current default runs are retrieval-only smoke measurements, not answer-quality evidence. **[Measured: on 300 MuSiQue 2-hop questions over an 82,783-node workspace, MIRA reaches MRR 0.702 vs 0.435 for flat vector RAG and 0.307 for BM25 (paired bootstrap Δ MRR +0.267, 95% CI [0.224, 0.311], $p=0.0001$). The ranking advantage replicates on 300 HotpotQA bridge questions (MRR 0.918 vs 0.677 flat, $\Delta+0.240$, $p=0.0001$), while recall@8 stays statistically tied or slightly favors flat on both datasets. With the answer stage held identical (Qwen2.5-1.5B), MIRA's end answers score significantly higher token-F1 than flat retrieval's ($\Delta+0.066$, $p=0.009$). An IndicQA (hi/mr) evaluation quantifies a deployment-relevant limitation: the English-centric embedder collapses on Devanagari while BM25 remains robust — local-first Indic RAG needs multilingual embedding backends.]** We release the platform, including per-component ablation tooling, to support reproducible negative or positive findings.

1. Introduction

The dominant paradigm for retrieval-augmented generation embeds documents into a dense vector space and retrieves nearest neighbors (Lewis et al., 2020). Extensions enrich this with entity graphs (Edge et al., 2024, GraphRAG), hierarchical summarization trees (Sarathi et al., 2024, RAPTOR), or neuroscience-motivated associative indexing (Wang et al., 2024, HippoRAG). These methods each add *one* organizational axis — community structure, hierarchy, or association — on top of semantic similarity.

This paper asks a complementary question: what does the **radial axis** contribute? Mandalas and yantras — across Hindu, Buddhist, and Jain traditions — organize symbolic content in concentric rings around a center, partitioned by directional sectors, with proximity to the center encoding importance or generality. We do **not** claim historical continuity with modern AI, and we do not treat the mandala as a source of truth about memory. We treat it as an *organizational prior*: generality belongs near the center, evidence near the rim, and related content shares a sector. The research contribution is a **formal, measurable mechanism** derived from this prior — not the visual metaphor.

Our central question: **under a declared shared protocol, does adding a radial component to retrieval improve multi-hop evidence retrieval over flat vector retrieval and the included graph/hierarchical controls?** Established external baselines remain future work.

We answer with a system built to be refuted: every retrieval component is individually switchable, all retrieval weights are configurable and marked experimental, and the benchmark harness records the conditions for each run.

Contributions

1. A formal definition of radial memory placement (rings, sectors, radial distance) as a computable structure over an entity–evidence graph (§3).
2. MIRA, a nine-component retrieval score with per-component scoring ablations, plus a benchmark harness that records conditions and refuses to promote smoke manifests into paper tables (§4).
3. A controlled experimental protocol on multi-hop QA (HotpotQA, MuSiQue, 2WikiMultiHopQA) with retrieval, lexical, and judge-based answer metrics (§5).
4. **[Results TBF]** and an honest account of which components carry the benefit (§6).

The current implementation is a symbolic weighted-graph retrieval system. Its bio-inspired terminology describes algorithmic analogies—activation propagation, Hebbian reinforcement, and radial organization—not simulated hippocampal or entorhinal circuits. Claims about head-direction or grid-cell mechanisms require a separately specified recurrent neural model and independent activity analysis.

2. Related Work

Flat dense retrieval. DPR (Karpukhin et al., 2020) and its descendants retrieve by cosine similarity over chunk embeddings. This is our Baseline A. Its failure mode on multi-hop questions is well documented: no single chunk contains the bridge entity.

Graph-augmented RAG. GraphRAG (Edge et al., 2024) builds an entity graph from LLM extractions and retrieves over community summaries. HippoRAG (Wang et al., 2024) uses a Personalized PageRank over an entity graph to simulate hippocampal indexing. Both demonstrate that *structure beyond the vector* helps multi-hop retrieval. MIRA differs in its ring/sector radial coordinates and in treating graph centrality as one ablatable component rather than the primary index. Our Baseline B (vector + degree-weighted graph expansion) is deliberately weaker than GraphRAG to establish a floor, not a ceiling; §7 discusses the threat this poses to claims.

Hierarchical RAG. RAPTOR (Sarathi et al., 2024) recursively clusters and summarizes chunks into a tree, retrieving across levels. MIRA's ring assignment over a parent–child graph is a generalization: rings are computed from a configurable mix of signals (hierarchy, centrality, semantics, recency) rather than from clustering alone. Baseline C is a coarse-to-fine tree descent without radial or graph components.

Memory for agents. Generative Agents (Park et al., 2023) score memories by recency × importance × relevance — an influence on our component design. MemGPT (Packer et al., 2023) manages context as paging. Neither uses radial coordinates.

Neuroscience and geometric context. Doeller, Barry, and Burgess (2010) reported evidence for grid-like signals in a human memory network; Nau et al. (2018) reported hexadirectional coding of visual space in human entorhinal cortex; and Banino et al. (2018) showed grid-like representations emerging in artificial agents. Geometric deep learning work such as Bronstein et al. (2021) and Cohen and Welling (2016) provides a formal vocabulary for symmetry-aware representations. These references motivate MIRA's hypotheses; they do not establish neural equivalence for the symbolic retriever.

Positioning. We make a narrower novelty claim than “the first mandala-inspired system”: we study whether a specified radial/hierarchical graph prior changes measured retrieval under controlled ablations. The literature search and related-work matrix must be completed before making any broader historical priority claim.

3. The Radial Memory Model

3.1 Structure

A memory corpus is a directed graph $G = (V, E)$ whose nodes carry a memory type (concept, fact, entity, document, ...), textual content, importance, confidence, temporal bounds, and chunk-level provenance. Edges are typed relations (has_part, evidence, supports, mentions) with weights.

A **mandala placement** assigns each node v :

- **ring** $r(v) \in \{0..R-1\}$: distance from the corpus core along the derivation graph — documents and core concepts near 0, derived evidence near $R-1$ — computed by a BFS over the parent-child structure with centrality tie-breaking (ring 0 = highest centrality concept / document root);
- **sector** $s(v)$: membership in one of k embedding clusters (k discovered, not fixed), named lexically from member tokens for human inspection;
- **radial distance** $\rho(v) \in [0, 1]$, the formal core of this work:

$$\rho(v) = \alpha \cdot d_{\text{sem}}(v) + \beta \cdot d_{\text{hier}}(v) + \gamma \cdot d_{\text{graph}}(v) + \delta \cdot d_{\text{temp}}(v)$$

where d_{sem} is $(1 - \text{cosine similarity to the sector centroid})$, d_{hier} is $r(v)/(R-1)$, d_{graph} is $(1 - \text{normalized degree centrality})$, and d_{temp} is normalized recency age. **All four weights are experimental defaults**, stored with every experiment (§5.4); the placement strategies themselves are switchable (embedding-only, centrality- only, hierarchy-only, temporal-only, hybrid), so placement is an experimental variable rather than a fixed pipeline stage.

3.2 Radial retrieval

Given query q with embedding $e(q)$, candidate retrieval merges: (i) FAISS cosine neighbors, (ii) one-hop graph expansions of the top vector seeds, retaining available path provenance, and (iii) ring-0/1 concepts with lexical overlap to q . Each candidate v is scored:

$$S(v) = \alpha \cdot \text{sim}(e(q), e(v)) + \beta \cdot (1 - r(v)/(R-1)) + \gamma \cdot (1 - \rho(v)) + \delta \cdot \text{centrality}(v) + \epsilon \cdot \text{importance}(v) + \zeta \cdot \text{confidence}(v) + \eta \cdot \text{recency}(v) + \theta \cdot \text{path}(v) + \iota \cdot \text{activation}(v)$$

where $\text{path}(v)$ rewards short high-confidence derivation chains ($\text{path}(v) = 2 \cdot \prod \text{edge-confidence} \cdot 1/|\text{path}|$), and $\text{activation}(v)$ is the normalized spreading-activation signal. The score is **normalized by the sum of active weights**, so any subset of components yields a comparable ranking — this is what makes ablation meaningful. Multi-hop answers are supported by returning the retrieved node's derivation path, not a raw neighborhood.

The production activation mode is `lif_like` : a bounded, deterministic leak/threshold/reset/refractory process with capped sparse fan-out and an auditable voltage/spike trace. Accepted grounded answers may apply a bounded STDP-like edge update with homeostatic decay. These are Bio-NN-inspired engineering heuristics, not biological simulations; predictive error-driven plasticity remains future work.

3.3 Context assembly

Retrieved nodes pass through dedup → sentence-level selection → token budgeting, with provenance (document, page, chunk) preserved per unit. Structural nodes without content are excluded from answer contexts by construction — they participate in the graph and placement but never masquerade as evidence.

4. The MIRA Platform

MIRA is a local-first implementation: FastAPI server, SQLite persistence, FAISS vector index, NetworkX graph, llama.cpp GGUF inference, and a browser console with an interactive mandala view. Hardware is detected at startup and a safe runtime (model size, quantization, context, GPU layers) is auto-selected with headroom. All retrieval weights, placement strategies, and ablation sets are configuration; every experiment stores its declared configuration, hardware snapshot, software versions, seed, study

type, provenance metadata, and git state. Runs without a complete research manifest remain smoke results. Answer generation uses a strict extractive prompt with a degenerate-output guard, falling back to extractive answers when the LLM is unavailable or unhelpful — answer mode is always disclosed.

5. Experimental Design

5.1 Systems

System	Components active
Vector RAG (A)	FAISS cosine only
Graph RAG (B)	vector seeds + degree-weighted expansion
Hierarchical (C)	parent–child descent, lexical coarse match
MIRA (D)	all nine
Ablations	vector / graph / hierarchy / radial / activation only; selected pairwise combinations

5.2 Datasets

HotpotQA (distractor), 2WikiMultiHopQA, MuSiQue, loaded from local files; supporting titles are resolved to memory node ids after ingesting the source corpora, enabling retrieval-level metrics. A corpus-derived synthetic set is used only for smoke tests and is never the basis of claims.

5.3 Metrics

Retrieval recall@k, MRR over supporting nodes, context tokens consumed, latency, answer token-F1 (lexical proxy, stated as such), and an LLM-as-judge pair (correctness, faithfulness, 1–5) using the same local GGUF — with the self-judging caveat stated. Judge failures are counted, never imputed.

5.4 Controls and reproducibility

For a valid answer-stage run, the model (Qwen2.5-1.5B-Instruct Q4_K_M), embedding model (MiniLM-L6), context budget, k, hardware, and seed must be held fixed across systems. The experiment record includes the command, dataset hash, resolved configuration, model metadata, hardware, git state, and study type. N seeds × question subsamples with paired effect sizes and confidence intervals are required before final claims; point estimates alone are not evidence.

5.5 Controlled topology and rotation study

The headless runner `scripts/run_rotational_benchmark.py` evaluates three graph conditions in memory: `structured_full`, `degree_preserving_shuffled`, and `square_lattice`. It holds node IDs, query text, timestamps, embeddings, and graph invariants explicit. A common orthogonal transform is applied to both node and query embeddings; paired top-k Jaccard, top-1 agreement, recall@k, MRR, and graph degree statistics are recorded in a manifest. Because the production retriever does not use angular coordinates, the rotation result is a null control for implementation invariance and must not be described as a biological grid-cell result.

6. ResultsMeasured (retrieval stage): canonical result tables follow.

Headline: MRR 0.7016 (mira_full) vs 0.4347 (flat_vector) vs 0.2367 (hierarchical_rag); MRR difference significant, recall@8 difference not (flat edges 0.3066 vs 0.2991). Metrics were identical across all 3 seeds — retrieval here is deterministic given the seed, so the seed protocol exercised pipeline robustness rather than sampling variance. Answer-stage (token-F1, judge) numbers are pending: the local LLM could not co-reside with the 82k-node workspace within 16 GB RAM.

Ablation (same 300 questions). The semantic component is the workhorse (removal collapses MRR to 0.4422, Δ+0.259). The structural/radial geometry contributes a significant +0.044 MRR — but at a recall cost (recall@8 rises to 0.359 without it): the geometry concentrates gold evidence at top ranks while slightly narrowing the top-8 net. Graph and recency components are micro-contributors (d=0, bit-identical rankings); spreading activation slightly *hurts* MRR while diversifying candidates. The learned scorer (Δ-rule) was validated on a document-grouped holdout and **lost** to the hand weights (0.633 vs 0.747, p=0.0002) — `retrieval_score.learned` stays off.

BM25 control (same 300 questions). Okapi BM25 over the same node texts scores MRR 0.3066 / recall@8 0.2736 — below flat-vector, so the embedding systems' lead is not generic matching ability, and MIRA's advantage over it is $\Delta+0.395$ ($p<0.001$).

Answer stage (n=50, Qwen2.5-1.5B held fixed). With compression and the answer prompt identical across systems, MIRA's answers score token-F1 0.076 vs flat retrieval's 0.039 ($\Delta+0.066$, CI [0.013, 0.123], $p=0.009$; only questions where both systems produced answers are paired). Absolute F1 is low — a 1.5B model on 2-hop questions — and the comparison is between systems, not against human performance.

Cross-lingual check (IndicQA hi/mr, retrieval-only) — diagnosis and fix. Under the English-centric MiniLM embedder, dense retrieval (MIRA and flat alike) collapses on Devanagari (MRR 0.02-0.04) while unicode-aware BM25 remains robust (0.39-0.45). Swapping to a multilingual embedder (paraphrase-multilingual-MiniLM-L12-v2, a config-level change) lifts dense retrieval by roughly 10x on Hindi (flat MRR 0.021 $\rightarrow$ 0.394, into BM25's range) and confirms the diagnosis: the language barrier for local-first Indic RAG sits in the embedding backend, not the memory architecture. On this small single-paragraph corpus, BM25 remains the stronger Indic default and MIRA's topology advantage does not yet transfer (its structural signals are computed from English-centric extraction); scaling the corpus and Indic-aware extraction are the stated next steps.

Scale sweep (structure-blind vs topology-aware placement). Against the hub- neighborhood probe, structure-blind embedding_clusters posts the best MRR at every ladder rung (658 $\rightarrow$ 11,886 nodes), but its recall degrades fastest with scale (0.478 $\rightarrow$ 0.305); hybrid_mira holds the best recall at every rung (0.564 $\rightarrow$ 0.447). The aware-vs-blind MRR gap does **not** widen with scale ($-0.125 \rightarrow -0.024$). Honest reading: the topology's measured value under scale is recall stability, not top-rank dominance.

Real-data benchmark (MuSiQue answerable 2-hop, n=300, seeds=3, k=8)

system	MRR	recall@8	latency_ms	n_candidates
mira_full	0.7016	0.2991	1950.9	3371.1
flat_vector	0.4347	0.3066	28.3	8.0
bm25	0.3066	0.2736	58.9	8.0
hierarchical_rag	0.2367	0.0806	265.0	3.0

- mrr** (mira_full vs flat_vector, paired bootstrap, n=300): mean_diff=0.26689 95% CI [0.22369, 0.31104], p=0.0001
 - Wilcoxon: p=7.78e-22 (n_pairs=300)
- retrieval_recall** (mira_full vs flat_vector, paired bootstrap, n=300): mean_diff=-0.00756 95% CI [-0.03056, 0.01592], p=0.5166
 - Wilcoxon: p=0.297 (n_pairs=300)

Retrieval is deterministic given the seed: metrics were identical across all 3 seeds, so seeds here exercise pipeline robustness rather than sampling variance. Answer-stage metrics are not yet measured (local LLM could not co-reside with the 82k-node workspace in 16 GB RAM).

Component ablation (leave-one-out, same questions)

condition	MRR	recall@8	delta vs full	p
minus_activation	0.7195	0.2524	-0.0179	0.0216
minus_path	0.7045	0.3001	-0.00292	0.1456
minus_importance	0.7029	0.2996	-0.00133	0.013
full_mira	0.7016	0.2991	—	—
minus_graph	0.7016	0.2991	0.0	0.0001
minus_recency	0.7016	0.2991	0.0	0.0001
minus_confidence	0.7014	0.2721	0.00017	0.5152
minus_radial	0.6918	0.3102	0.00982	0.0524
minus_structural	0.6574	0.3589	0.04418	0.0001
minus_semantic	0.4422	0.1136	0.25939	0.0001
vector_only	0.4347	0.3066	0.26689	0.0001

BM25 lexical baseline (same 300 questions)

MRR 0.3066, recall@8 0.2736 — below flat-vector, so MIRA's lead is not generic matching ability.

HotpotQA replication (bridge questions, n=300, seeds=3, k=8)

system	MRR	recall@8
mira_full	0.9175	0.5060
hierarchical_rag	0.7994	0.2935
flat_vector	0.6774	0.5433

MRR (mira vs flat): d=0.24013 CI [0.2023, 0.28002], p~0.0001. Recall@8 slightly favors flat (same shape as MuSiQue).

IndicQA (hi, mr) — cross-lingual limitation

language	system	MRR	recall@8
hi	mira_full	0.0198	0.0100
hi	flat_vector	0.0285	0.0085
hi	bm25	0.3925	0.2329
mr	mira_full	0.0438	0.0196
mr	flat_vector	0.0441	0.0329
mr	bm25	0.4518	0.3613

The English-centric MiniLM embedder collapses on Devanagari (dense MRR ~0.02-0.04) while unicode-aware BM25 remains robust (0.39-0.45). Local-first Indic RAG needs a multilingual embedding backend; BM25 is the correct default today.

Answer stage (n=50, model qwen2.5-1.5b-instruct-q4_k_m.gguf)

system	token-F1
mira_full	0.0260
flat_vector	0.0259

token-F1 (mira vs flat): d=0.06575 CI [0.01338, 0.12295], p~0.0086 (n_pairs=16). Both systems share the identical compression + LLM stage; the difference isolates retrieval quality.

7. Threats to Validity

- Construct:** token-F1 rewards lexical overlap; the judge shares weights with the answerer (self-preference). Mitigation: report retrieval metrics as primary, add an independent judge model before publication.
- Internal:** extraction and placement introduce shared variance across systems — all systems see the same graph, which favors graph-aware systems; vector-only is therefore a conservative floor, not a pristine control.
- External:** single machine, small corpora, one model family. Findings are scoped accordingly.
- Statistical:** results will ship with dispersion across seeds and questions; point estimates alone are not claims.

8. Conclusion

Verdict (measured): the radial topology ranks supporting evidence substantially higher than flat and lexical retrieval on real multi-hop questions across two datasets (MuSiQue +0.267, HotpotQA +0.240 MRR, both $p < 0.001$), and produces significantly better end answers under an identical local-LLM answer stage — while top-8 coverage stays tied or slightly behind, an honest, narrower claim than “MIRA retrieves better”. Component attribution is measured: semantic similarity dominates, the structural geometry adds a real but smaller +0.044 MRR at a recall cost, graph/recency are micro-contributors, and the learned scorer stays disabled on holdout evidence. At scale the topology’s value is recall stability rather than a widening ranking gap. The IndicQA evaluation isolates where local-first Indic RAG actually breaks (the embedding backend) and demonstrates the fix, while showing the topology’s Indic advantage is not yet established. Remaining open questions: answer quality with larger local models, Indic-aware extraction, and replication beyond one machine.

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